

India's Geographic Vulnerability to Climate Change

INDUSTRY UPDATES

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Manasvi Patel



World Environmental Health Day 2024

World Environmental Health Day 2024 focuses on the theme "Environmental Health: Creating Resilient Communities through Disaster Risk Reduction and Climate Change Mitigation and Adaptation." This theme underscores the critical need to enhance community resilience in the face of climate-related disasters, which have become increasingly frequent and severe due to global warming. By emphasizing disaster risk reduction (DRR) and climate change adaptation, this year's theme calls for integrated approaches that not only protect public health but also ensure sustainable development.

India's unique geography and climate make it one of the most disaster-prone countries in the world. With 27 out of 29 states and 7 union territories susceptible to natural hazards like floods, cyclones, droughts, and earthquakes, the country faces an uphill battle against the devastating impacts of climate

change. As World Environmental Health Day 2024 approaches, it's crucial for India to take bold steps to enhance community resilience and protect its vulnerable populations.

Geographical Factors Contributing to Disaster Risk

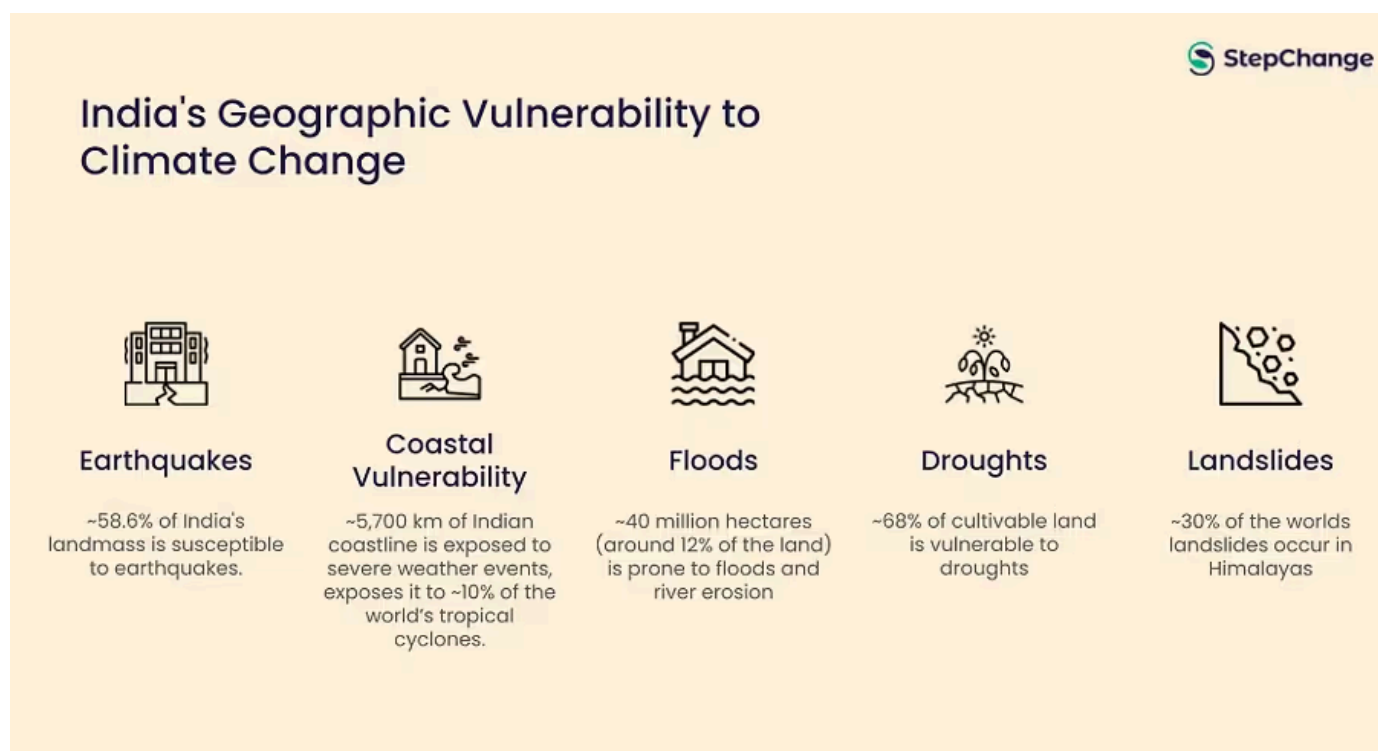
India's geographical factors significantly contribute to its disaster risk profile, making it one of the most disaster-prone countries in the world. The combination of its diverse topography, climatic conditions, and demographic factors creates a complex landscape of vulnerabilities:

Earthquakes: Approximately 58.6% of India's landmass is susceptible to earthquakes, with the country situated in a seismically active zone due to the collision of the Indian and Eurasian tectonic plates.

Coastal Vulnerability: India's coastline of 7,516 km, with approximately 5,700 km exposed to severe weather events, exposes it to nearly 10% of the world's tropical cyclones.

Floods and **droughts**: about 40 million hectares (around 12% of the land) are prone to floods and river erosion, while 68% of cultivable land is vulnerable to droughts.

Landslides and avalanches: The hilly regions are also at risk from landslides and avalanches, further compounding the disaster risk faced by various communities across the nation.



Seismic Activity

India is situated at the convergence of several tectonic plates, making it prime for earthquakes. ~58% of the country's landmass is classified as being at risk for earthquakes of moderate to very high intensity. This seismic vulnerability is particularly pronounced in regions such as the Himalayan belt, which experiences tremors due to ongoing tectonic activity.

The Gujarat (Bhuj) Earthquake (2001)

For instance, the [2001 Bhuj earthquake](#), which registered a magnitude of 7.7, resulted in over 20,000 deaths and injured more than 150,000 others. The earthquake affected approximately 28 million people, destroyed more than 400,000 buildings, with the telecom building collapsing, Bhuj was isolated from the rest of the state; the most seriously affected was the Kandla port (the busiest port in India) which caters to handling crucial imports of petroleum products, crude oil and chemicals and export of agricultural commodities, with about five of the 10 jetties damaged. Over 10,000 small and medium industrial units had to stop production due to damage to plants, factories, and machinery. The Indian State Department [estimates](#) that the earthquake affected, directly or indirectly, 15.9 million people out of 37.8 million. The government estimated the direct economic loss at \$1.3 billion, while estimates indicate this could be as high as \$5 billion.

Following the earthquake, emergency relief operations were launched by both government and non-governmental organizations (NGOs):

[ActionAid India](#) mobilized local community-based organizations to provide immediate assistance in the most affected areas, by establishing temporary shelters, distributed food and medical supplies, and set up psycho-social counseling services for survivors.

The government initiated a [comprehensive rehabilitation plan](#) that included new building codes to ensure that the structures were designed to be earthquake-resistant; and infrastructure redevelopment of roads, schools, hospitals, and utilities was prioritized.

Emphasizing on community participation in planning and implementing disaster management strategies, village-level committees were formed to enhance local capacity for disaster response.

A [State Disaster Management Plan](#) was created that integrated lessons learned from the earthquake, and focused on risk assessment, early warning systems, and community-based disaster management practices aimed at reducing vulnerability to future seismic events.



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Earthquakes

~58.6% of India's landmass is susceptible to earthquakes.

The Gujarat Earthquake, 2001

- **Devastating impact:** The 2001 Bhuj earthquake impacted 15.9 million people, with estimates indicating a cost of \$5 Billion.
- **Relief and rehabilitation:** Both government and NGOs provided immediate relief and implemented long-term rehabilitation plans.
- **Community involvement:** Village-level committees were formed to enhance local capacity for disaster response.
- **Lessons learned:** The earthquake led to the creation of a State Disaster Management Plan aimed at reducing future vulnerability.

The resilience-building activities undertaken after the Gujarat earthquake reflect a comprehensive approach that combines immediate relief with long-term strategies aimed at enhancing community preparedness and reducing vulnerability to future disasters. These efforts underscore the importance of integrating disaster risk reduction into development planning to safeguard lives and livelihoods in seismically active regions like India

Coastal Vulnerability

India's long coastline of 7,516 km, with approximately 5,700 km exposed to severe weather events, makes it highly vulnerable to cyclones and tsunamis, [exposing](#) it to nearly 10% of the world's tropical cyclones. This vulnerability is particularly pronounced in states like Odisha and Andhra Pradesh, which have borne the brunt of devastating cyclones in recent years.

Cyclone Fani (2019)

Cyclone Fani, which made landfall in Odisha on May 3, 2019, serves as a prime example of the destruction that coastal areas face. The [extremely severe](#) cyclonic storm caused extensive damage across 14 districts in Odisha,

affecting at least 10 million people in the North-eastern States of India. Despite the early evacuation of a million people, the cyclone has severely affected the lives and livelihoods of more than 28 million people across 3 States in India. Total damage to existing stock of assets has been estimated at over \$2.4 billion, whereas the economic losses flowing from the disaster are estimated at nearly ~\$1.1 billion.

In response to the devastation caused by Cyclone Fani, several resilience-building activities were implemented to enhance disaster preparedness and recovery efforts:

The Government of Odisha coordinated one of the largest emergency preparedness exercises, mobilizing over 45,000 volunteers and 100,000 officials to ensure effective communication and evacuation protocols. This effort enabled the successful evacuation of approximately 1.47 million people from low-lying areas before the cyclone made landfall.

The state government constructed hundreds of cyclone shelters along the coast to provide safe refuge during storms, which were stocked with essentials like food and water.

Local stakeholders, NGOs, and civil society organizations collaborated with the government to conduct training programs on disaster preparedness including educating communities about evacuation routes and safety measures

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Coastal Vulnerability

~5,700 km of Indian coastline is exposed to severe weather events, exposes it to ~10% of the world's tropical cyclones.

Cyclon Fani, Odisha, 2009

- **Devastating impact:** Cyclon Fani impacted 28 million people across 3 states, with estimates indicating a loss of \$2.4 Billion.
- **Early evacuation:** The government coordinated a large-scale evacuation effort, successfully relocating over 1.47 million people.
- **Infrastructure and Shelter Initiatives:** Constructed cyclone shelters stocked with essential supplies to provide safe refuge during storms, and for disaster preparedness
- **Lessons learned:** Collaborated effort between local stakeholders and the government to conduct training programs on disaster preparedness to enhance community resilience

The resilience-building activities undertaken after Cyclone Fani reflect a comprehensive approach that combines immediate response efforts with long-term strategies aimed at enhancing community preparedness and reducing vulnerability to future disasters. The lessons learned from this cyclone underscore the importance of integrating disaster risk reduction into development planning to safeguard lives and livelihoods in coastal regions like Odisha.

Flood-Prone Areas

Over 40 million hectares (about 12% of India's land) is [prone to floods and river erosion](#). The monsoon season often brings heavy rainfall leading to river overflow and flash floods, particularly in states like Assam and Bihar.

Assam Floods (2023)

Assam, a state in northeastern India, is characterized by its fertile Brahmaputra Valley, which supports a rich agricultural sector. The state's agriculture is primarily based on rice, tea, and jute, and it benefits from the regular monsoon rains that sustain its crops. The geographical features of Assam, including its river basins and low-lying areas, contribute significantly to its susceptibility to flooding. In 2023, Assam experienced devastating floods that [impacted](#) over 2.5 million people.

In response to the challenges posed by flooding, particularly following the severe floods in 2023, several resilience-building activities have been undertaken in Assam:

The Assam government allocated Rs 1200 crs in 2023-24 for the [construction and maintenance](#) of flood control infrastructure, including embankments and drainage systems and investment aimed to enhance the state's capacity to manage floodwaters effectively and reduce the risk of future inundations.

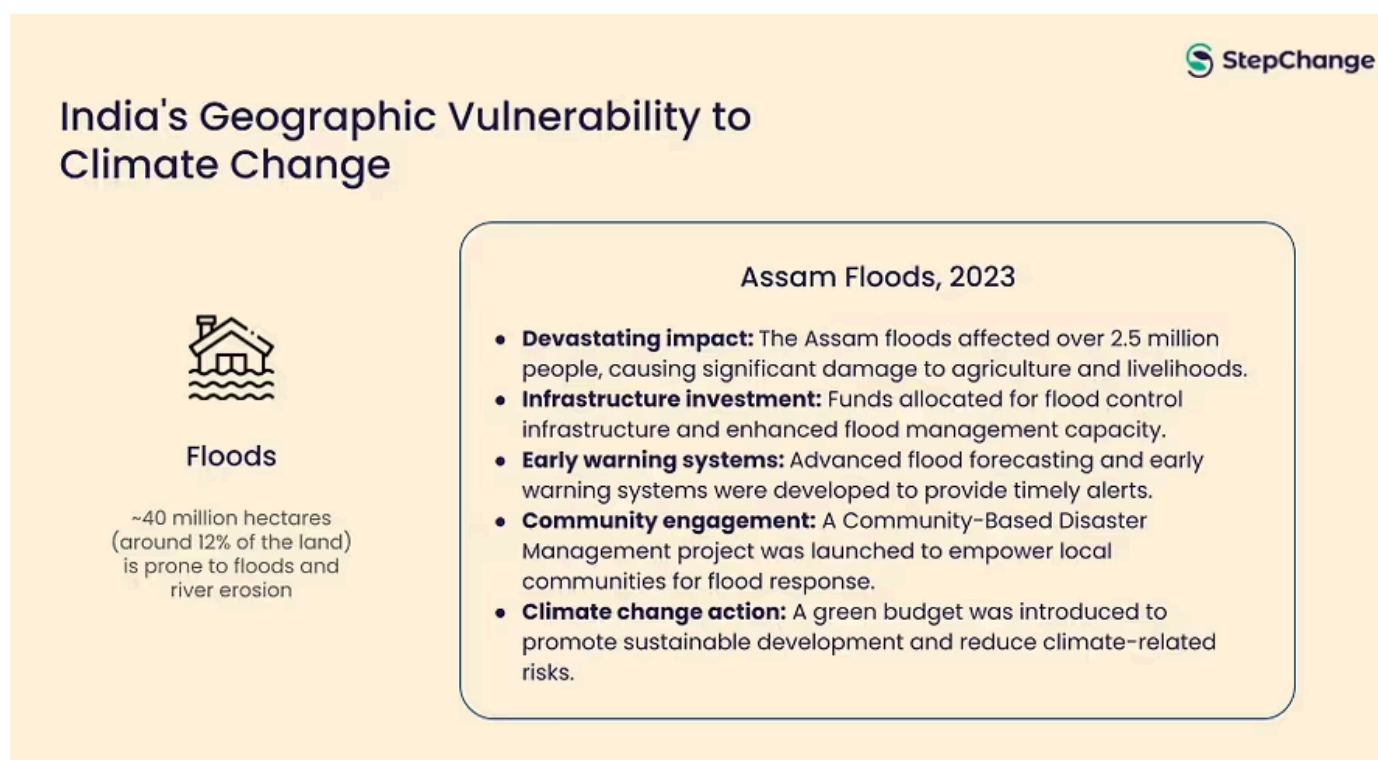
The Assam State Disaster Management Authority (ASDMA) invested ₹25 crores in developing advanced flood forecasting and early warning systems. These systems provide alerts up to three days in advance, allowing communities to prepare for impending floods and evacuate if necessary.

The Pradhan Mantri Fasal Bima Yojana (PMFBY) [covers](#) around 56% of farmers in Assam as of 2023, providing insurance against crop losses due to

natural calamities. This scheme is crucial for protecting farmers' livelihoods and ensuring financial support during disasters.

A Community-Based Disaster Management project was launched in 2023, aimed at building local capacity for flood management in 100 selected gram panchayats focusing on empowering communities with knowledge and resources to respond effectively to floods.

In 2023, Assam introduced a '[green budget](#)' committing approximately \$2 billion towards climate change action and disaster risk reduction, aiming to enhance long-term resilience against climate-related disasters through sustainable development practices.



India's Geographic Vulnerability to Climate Change



Floods

~40 million hectares
(around 12% of the land)
is prone to floods and
river erosion

Assam Floods, 2023

- **Devastating impact:** The Assam floods affected over 2.5 million people, causing significant damage to agriculture and livelihoods.
- **Infrastructure investment:** Funds allocated for flood control infrastructure and enhanced flood management capacity.
- **Early warning systems:** Advanced flood forecasting and early warning systems were developed to provide timely alerts.
- **Community engagement:** A Community-Based Disaster Management project was launched to empower local communities for flood response.
- **Climate change action:** A green budget was introduced to promote sustainable development and reduce climate-related risks.

The resilience-building activities implemented in Assam following recent floods reflect a proactive approach to disaster risk reduction. By investing in infrastructure, enhancing early warning systems, promoting community engagement, and introducing innovative financial mechanisms like insurance schemes and green budgets, Assam is working towards building a more resilient agricultural sector capable of withstanding the impacts of flooding.

Drought and Water Scarcity

Drought and water scarcity are significant challenges in India, with [about](#) 68% of the country's cultivable area vulnerable to these conditions.

Rajasthan Drought and Water Scarcity

Rajasthan, the largest state in India, faces significant challenges related to water scarcity and drought leading to crop failures and economic distress among farmers. The state has [limited water resources](#), with only 1.16% of India's total water supply, making it particularly susceptible to prolonged drought conditions. It's among the most [vulnerable to droughts](#), with 98% of its 250 village blocks in sectors marked as "dark zones" – areas with dangerously low groundwater levels – and almost 7% of the land uncultivable.

In response to the recurring drought conditions, several resilience-building activities have been implemented in Rajasthan:

Rainwater harvesting and water conservation techniques were introduced to mitigate the effects of droughts and were prioritized via traditional techniques such as Anicuts, Bais, and tanks. In 2023, after an [unusually high rainfall](#) of 588.9 mm (nearly 2X the normal amount) villagers utilized these methods to direct the water into ponds and storage system, providing drinking water [for upto](#) 1.5 years in some villages.

The Rajasthan Water Sector Restructuring Project, costing a total of \$495 million, of which \$345 million is [financed](#) by the New Development Bank, aimed to rehabilitate the Indira Gandhi Canal System, covering over 990 km of canal systems, reclamation of more than 33,000 km of waterlogged areas, and implementation of micro-irrigation techniques.

Government schemes like the [Pradhan Mantri Krishi Sinchai Yojana](#) (PMKSY) were implemented to focus on enhancing irrigation facilities, along with the introduction of a crop insurance scheme to protect farmers from losses due to drought.

India's Geographic Vulnerability to Climate Change



Droughts

~68% of cultivable land is vulnerable to droughts

Rajasthan

- **Devastating impact:** Rajasthan's limited water resources and arid climate make it highly vulnerable to droughts, leading to crop failures and economic distress.
- **Water conservation:** Rainwater harvesting and water conservation techniques were introduced to mitigate the effects of droughts.
- **Infrastructure development:** The Rajasthan Water Sector Restructuring Project aimed to rehabilitate irrigation systems and improve water management.
- **Govt. Financial support:** Government schemes like PMKSY and crop insurance were implemented to protect farmers from losses due to drought.

Despite these resilience-building efforts, Rajasthan continues to face the challenges posed by recurring droughts. The state's vulnerability is exacerbated by its arid climate, limited water resources, and high dependence on agriculture. Sustained efforts and investments in water management, and agricultural practices are crucial for mitigating the impact of droughts and ensuring food security in the region.

Topography and Landslides

India's diverse topography includes mountainous regions that are particularly vulnerable to landslides and avalanches, especially during the monsoon season. The Himalayan region, along with the Western Ghats, experiences heavy rainfall that can destabilize slopes and trigger landslides. One of the most devastating recent events occurred in Wayanad, Kerala, where a series of landslides on July 30, 2024, resulted in significant loss of life and property.

The Wayanad Landslides (2024)

On July 30, 2024, Wayanad district in Kerala was struck by catastrophic landslides that resulted from extreme rainfall. The region received 572 mm of rainfall within 48 hours leading up to the disaster, causing hillsides to collapse in the villages of Mundakkai and Chooralmala. This resulted in over 336 fatalities, ~397 were injured, and at least 118 missing. The debris flow altered the course of the Iruvanjipuzha River, exacerbating flooding conditions in surrounding

areas. The National Remote Sensing Centre (NRSC) [reported](#) that about 86,000 square meters of land had shifted due to the landslide.

In response to the Wanayd landslides, several activities were initiated:

A [rescue operation](#) was mobilized involving the Indian Army, National Disaster Response Force (NDRF) and local authorities, over 9,300 people were accommodated in relief camps across the district.

[Experts](#) have called for implementing site-specific land-use policies and improving drainage management to mitigate future risks. Additionally, establishing local landslide early warning systems (LEWS) has been recommended to enhance community preparedness.

The Wayanad landslides highlight the urgent need for improved land management practices and disaster preparedness in vulnerable regions of India. By implementing effective resilience-building strategies, including community engagement and infrastructure improvements, it is possible to mitigate the impacts of future landslides and protect lives and livelihoods.



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Landslides

~30% of the worlds
landslides occur in
Himalayas

Landslides in Wayanad, 2024

- **Devastating impact:** The Wayanad landslides caused significant loss of life and property, with over 336 fatalities and widespread damage.
- **Rescue and relief:** A large-scale rescue operation was mobilized to provide assistance to affected communities.
- **Community preparedness:** Experts recommended implementing land-use policies, improving drainage management, and establishing local landslide early warning systems.
- **Resilience building:** The landslides highlight the need for effective land management practices and disaster preparedness to mitigate future risks.

India's unique geographical factors significantly contribute to its vulnerability to natural disasters. The country's diverse topography, climatic conditions, and demographic factors create a complex landscape of risks, including earthquakes, coastal vulnerability, floods, droughts, landslides, and avalanches.

These vulnerabilities are exacerbated by the increasing frequency and intensity of climate-related events due to global warming.

To address these challenges, India must adopt a comprehensive approach that integrates disaster risk reduction and climate change adaptation into its development plans. Building resilient communities requires a multi-faceted strategy that includes strengthening early warning systems, investing in resilient infrastructure, promoting community engagement, and adopting sustainable practices.

The second part - Building a Climate Resilient India: A Call for Action, coming soon!

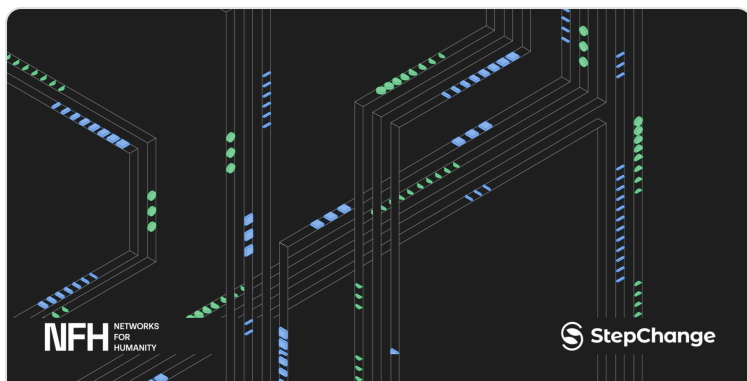
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